

I. Introduction

Purpose of the Guide

Welcome to the comprehensive guide for the Elite Options Trading System. This document is designed to provide you with a thorough understanding of all the signals, metrics, and visualizations generated by the system. Our goal is to empower you to interpret this information effectively and integrate it into your trading strategies to make more informed decisions.

Overview of the Trading System and its Core Philosophy

The Elite Options Trading System is built upon the philosophy that inefficiencies and predictable patterns arise from the hedging activities of market makers and significant options order flow. By analyzing complex metrics derived from options market data (such as Gamma Exposure (GEX), Delta Exposure (DEX), and various flow indicators), the system aims to identify key price levels, potential market turning points, and periods of stability or volatility.

The system processes raw options data, calculates a suite of proprietary and standard metrics (including MSPI, SDAG, TDPI, VRI, SAI, SSI, CFI), generates actionable trading signals, and provides rich visualizations through a dedicated dashboard and standalone chart outputs. It is highly configurable, allowing you to tailor its sensitivity and focus to your specific trading style and market outlook.

How to Use This Guide

This guide is structured to build your understanding progressively:

1. **Core Concepts & Terminology:** Familiarize yourself with fundamental options trading terms relevant to the system.
2. **Individual Metrics Explained:** Deep dive into each specific metric – what it measures, how it's calculated (conceptually), how to interpret it, and its configuration.
3. **Trading Signals Explained:** Understand each discrete trading signal the system generates – its triggers, interpretation, and practical use.
4. **Cohesive Analysis:** Learn how to synthesize multiple metrics and signals for a holistic market view and develop trading plans.
5. **Visual Guide:** Get acquainted with the dashboard layout and how to interpret the various charts and tables.

6. **Advanced Configuration:** Explore how to customize the system via `config_v2.json`.
7. **Troubleshooting & FAQ:** Find answers to common questions and issues.
8. **Glossary:** A quick reference for all terms.

We recommend reading through the guide sequentially first. Later, you can use it as a reference. Pay close attention to the examples and how configuration settings can impact the output.

II. Core Concepts & Terminology

Understanding the following terms is essential for effectively using the system:

- **Options Greeks:** These measure the sensitivity of an option's price to various factors.
 - **Delta:** Measures the rate of change of an option's price relative to a \$1 change in the underlying asset's price. Ranges from 0 to 1 for calls, -1 to 0 for puts.
 - **Gamma:** Measures the rate of change of an option's Delta relative to a \$1 change in the underlying asset's price. Highest for at-the-money options near expiration.
 - **Theta:** Measures the rate of change of an option's price relative to the passage of time (time decay). Generally negative for long options.
 - **Vega:** Measures the rate of change of an option's price relative to a 1% change in the implied volatility of the underlying asset.
 - **Charm (Delta Decay):** Measures the rate of change of an option's Delta with respect to the passage of time.
 - **Vanna:** Measures the rate of change of an option's Delta with respect to a change in implied volatility. Also, the rate of change of Vega with respect to a change in the underlying price.
 - **Vomma:** Measures the rate of change of an option's Vega with respect to a change in implied volatility.
- **Open Interest (OI):** The total number of outstanding options contracts that have not been settled or closed. High OI at a strike can indicate significance.
- **Volume:** The total number of options contracts traded during a given period. High volume indicates activity and interest at certain strikes.
- **GEX (Gamma Exposure):** The total gamma sensitivity across all options at a particular strike price or for the entire market. Positive GEX tends to suppress volatility (dealers hedge by buying into dips and selling into rallies), while negative GEX (less common for broad indices) can exacerbate it. (Knowledge: User_1)
 - **Skew-Adjusted GEX:** GEX that accounts for volatility skew, providing a more realistic measure of hedging impact. (Knowledge: User_6, User_8)

- **DEX (Delta Exposure):** The total delta sensitivity at a strike or for the market. Indicates the net directional hedging required by market makers.
- **Volatility Skew:** The difference in implied volatility (IV) between out-of-the-money (OTM), at-the-money (ATM), and in-the-money (ITM) options. Typically, OTM puts have higher IV than OTM calls for equities and indices.
- **"Gamma Wall":** A price level with a very large concentration of GEX, acting as strong support or resistance, often like a price magnet. (Knowledge: User_1, User_35, User_38, User_43)
- **"Volatility Trigger":** A price level, often identified by strongly negative SDAG values, where market volatility dynamics are expected to change significantly. Breaching these levels can lead to accelerated moves. (Knowledge: User_33, User_15, User_19)
- **Market Maker Hedging:** Options market makers aim to remain delta-neutral. As the underlying price moves, they must buy or sell the underlying asset to offset the changing delta of their options book. This hedging activity can influence price, especially around large GEX strikes.
- **Order Flow Imbalance (OFI):** The net difference between buying and selling pressure in the options market, often measured by volume or value of aggressive buy orders versus sell orders.

III. Individual Metrics Explained

This section delves into the individual metrics calculated by the system. Each metric provides a unique lens through which to analyze market structure and potential price movements. Understanding each one is crucial before attempting to synthesize them into a cohesive trading strategy.

1. Delta Adjusted Gamma Exposure (DAG - Custom)

- **Metric Name & Abbreviation:** Delta Adjusted Gamma Exposure (DAG - Custom)
- **Conceptual Explanation:** The custom DAG metric is a core component of the MSPI. It aims to measure the combined influence of gamma and delta exposures at various strike prices. Unlike raw GEX (Gamma Exposure) which only shows the magnitude of potential hedging, DAG incorporates the directional bias of Delta Exposure (DEX). This provides a more nuanced view of how market maker hedging might influence price, especially around significant strike levels. It tries to identify points where dealer hedging could either accelerate a move (if delta and gamma align) or dampen it (if they oppose).
- **Simplified Calculation Insight:** The custom DAG is calculated within the `IntegratedTradingSystem` by considering both the `gamma_exposure_source_col`

(typically `gxoi`) and `delta_exposure_source_col` (typically `dxoi`) from the input options data. The exact proprietary calculation logic within the `_calculate_dag` method of the original `integrated_strategies.py` (which `integrated_strategies_v2.py` builds upon) combines these, potentially applying coefficients based on their alignment (e.g., `dag_alpha` from `config_v2.json` which has `aligned`, `opposed`, `neutral` factors).

- **Visual Representation:** [Placeholder for DAG chart - typically a bar chart showing DAG values per strike price, similar to GEX/DEX charts but representing the combined metric].
- **Interpretation Guide:**
 - High positive DAG values at a strike might suggest a level where market maker hedging could support price or accelerate a move upwards if price is already moving towards it.
 - High negative DAG values (depending on how it's scaled, or if it represents net exposure) could indicate resistance or areas where hedging might accelerate a downward move.
 - The distribution of DAG across strikes helps identify key inflection points.
 - The interpretation is heavily influenced by the `dag_alpha` coefficients in `config_v2.json` -> `data_processor_settings` -> `coefficients`. If `aligned` is > 1 , it amplifies the effect when delta and gamma suggest the same hedging pressure.
- **Practical Use Cases & Examples:**
 - Identifying strong support/resistance levels based on combined dealer positioning.
 - Gauging the potential for price to be "pinned" to or repelled from certain strikes due to hedging flows.
- **Relationship to Other Metrics:** DAG is a primary input into the Market Structure Position Indicator (MSPI). It's also conceptually related to the various SDAG methodologies, which offer alternative ways to combine gamma and delta, often with skew adjustment.
- **Configuration Notes:**
 - `data_processor_settings.weights.*.dag_custom` : Weight of this custom DAG in the MSPI calculation, varies by time or IV context.
 - `data_processor_settings.coefficients.dag_alpha` : Modifies DAG based on delta/gamma alignment.
 - `strategy_settings.gamma_exposure_source_col` : Source column for gamma (e.g., `"gxoi"`).
 - `strategy_settings.delta_exposure_source_col` : Source column for delta (e.g., `"dxoi"`).

2. Skew and Delta Adjusted GEX (SDAG) Methodologies

The system implements several SDAG methodologies to provide a more refined view of market pressures by potentially incorporating volatility skew.

- **Metric Name & Abbreviation:** Skew and Delta Adjusted GEX (SDAG)
 - SDAG Multiplicative
 - SDAG Directional
 - SDAG Weighted
 - SDAG Volatility-Focused
- **Conceptual Explanation:** SDAG metrics build upon the concept of DAG by explicitly accounting for volatility skew (if `use_skew_adjusted_for_sdag` is true and `skew_adjusted_gamma_source_col` data is available) and then combining this with delta exposure. The goal is to get an even more accurate picture of true hedging pressures. Different methodologies combine these factors in slightly different ways, offering various perspectives.
 - **Knowledge User_10, User_11, User_15, User_16, User_19, User_21, User_22:** These knowledge items emphasize that SDAG provides a more comprehensive view than GEX or DEX alone, helps identify precise support/resistance, and that alignment across methodologies gives higher conviction.
- **Simplified Calculation Insight:**
 - Each SDAG methodology (`multiplicative` , `directional` , `weighted` , `volatility_focused`) has its own formula for combining gamma (potentially skew-adjusted) and delta. These are implemented in `_calculate_sdag_multiplicative` , `_calculate_sdag_directional` , `_calculate_sdag_weighted` , and `_calculate_sdag_volatility_focused` methods in `integrated_strategies_v2.py` .
 - **Multiplicative:** $\text{Gamma_Component} * (1 + \text{Delta_Component} * \text{delta_weight_factor})$
 - **Directional:** $\text{Gamma_Component} * \text{sign}(\text{Gamma_Component} * \text{Delta_Component}) * (1 + \text{abs}(\text{Delta_Component} * \text{delta_weight_factor}))$
(Note: `delta_weight_factor` is applied to Delta before `abs` in current code, effectively $(1 + \text{abs}(\text{Delta_Component}) * \text{delta_weight_factor})$ if `delta_weight_factor` is from config for the method, or just $(1 + \text{abs}(\text{Delta_Component}))$ if `delta_weight_factor` is not used as a multiplier for delta component itself in the formula. The `integrated_strategies_v2.py` code for `_calculate_sdag_directional` uses $(1 + \text{delta_component.abs}() * \text{methodology_config.get("delta_weight_factor", 0.5)})$ which seems to be a slight variation from pure $(1 + |\text{Delta}|)$ if `delta_weight_factor` is not 1. The knowledge base `user_21` and `user_22` suggests $(1 + |\text{Delta}|)$ or $(1 + \text{Delta} * \text{delta_weight_factor})$.)

sign(Skew_Adjusted_GEX)) for Volatility-Focused. The code uses `delta_weight_factor` from config for most methods.

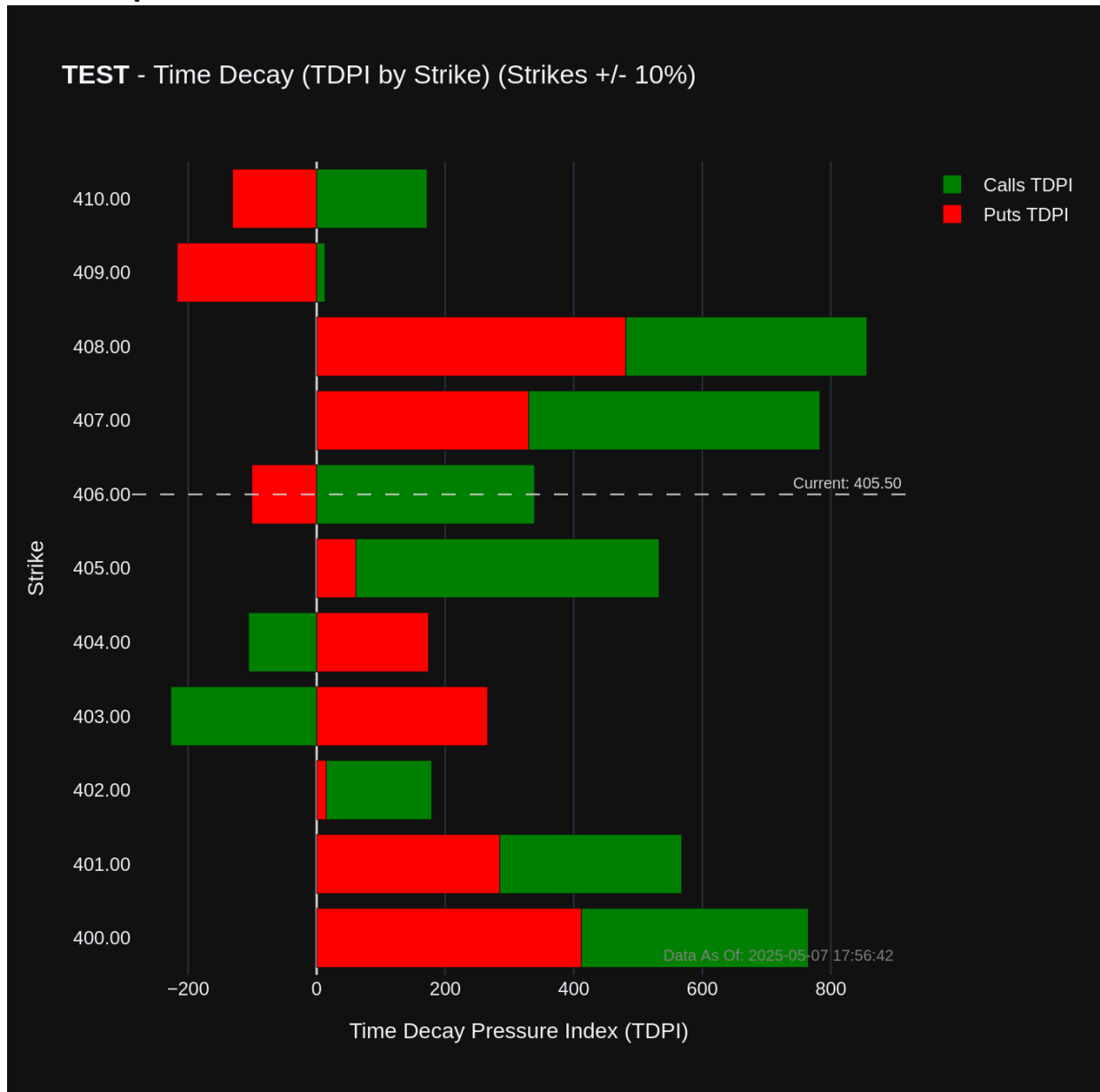
- **Weighted:** $(w1_gamma * \text{Gamma_Component} + w2_delta * \text{Delta_Component}) / (w1_gamma + w2_delta)$
- **Volatility-Focused:** $\text{Gamma_Component} * (1 + \text{Delta_Component} * \text{sign}(\text{Gamma_Component}) * \text{delta_weight_factor})$ (Code uses `delta_weight_factor` from config).
- The `Gamma_Component` is either standard GEX (e.g., `gxoi`) or Skew-Adjusted GEX (e.g., `sgxoi`) based on `use_skew_adjusted_for_sdag`.
- **Knowledge User_21, User_22:** Provide specific formulas for these methodologies.
- **Visual Representation:** [Placeholder for SDAG charts - typically line or bar charts per methodology, showing values per strike. A combined chart showing alignment is also powerful].
- **Interpretation Guide:**
 - **Knowledge User_15, User_16, User_19:** SDAG values > 1.5 indicate extremely strong positive signals (support/resistance), while < -1.5 are extremely strong negative signals (volatility triggers).
 - **Knowledge User_19:** Provides detailed ranges: >1.5 (extremely strong positive), 1.0-1.5 (strong positive), 0.5-1.0 (moderate positive), etc.
 - Declining slopes from peaks indicate decreasing stability.
 - **Knowledge User_10, User_15, User_16, User_21, User_22:** Alignment of multiple (or all four) SDAG methodologies at specific price levels provides the highest conviction trading signals.
 - The `sdag_conviction` signal is generated when a configurable majority of enabled SDAG methodologies agree.
- **Practical Use Cases & Examples:**
 - Identifying high-conviction support and resistance levels.
 - Pinpointing potential volatility trigger points.
 - Range trading between key SDAG levels.
 - **Knowledge User_21, User_22:** Suggests focusing on Multiplicative/Directional for day trading, Weighted for swing trading, and Volatility-Focused for options strategies.
- **Relationship to Other Metrics:** Enabled SDAG methodologies (if their `weight_in_msipi` > 0) are normalized and then included in the MSPI calculation. They provide an alternative or supplementary view to the custom DAG.
- **Configuration Notes:**
 - `strategy_settings.gamma_exposure_source_col` ,
`strategy_settings.delta_exposure_source_col` ,
`strategy_settings.skew_adjusted_gamma_source_col` .

- `strategy_settings.use_skew_adjusted_for_sdag` : Boolean to use skew-adjusted GEX for SDAGs.
- `strategy_settings.dag_methodologies.enabled` : List of which SDAGs to calculate.
- `strategy_settings.dag_methodologies.[method_name]` : Specific parameters for each SDAG (e.g., `delta_weight_factor` , `w1_gamma` , `w2_delta` , `weight_in_mspi`).
- `strategy_settings.sdag_conviction_threshold_pct` : Percentage of enabled SDAGs that must agree for a conviction signal.
- `data_processor_settings.weights.*.*.sdag_[method_name]_norm` : Weights of normalized SDAGs in MSPI if `weight_in_mspi` for the method is > 0.

3. Time Decay Pressure Indicator (TDPI)

- **Metric Name & Abbreviation:** Time Decay Pressure Indicator (TDPI)
- **Conceptual Explanation:** TDPI is designed to measure the potential market impact of time decay (theta) on options positions, particularly as expiration approaches. It considers how accelerating time decay might force hedging adjustments or create pinning effects around strikes with large open interest, especially those near the current underlying price.
- **Simplified Calculation Insight:** TDPI is calculated in `_calculate_tdpi` . It typically uses a Gaussian function centered around the current underlying price, weighted by open interest or gamma at different strikes. The width of this Gaussian (`tdpi_gaussian_width`) determines how far out the "pressure" is felt. It also incorporates an ATR (Average True Range) concept, potentially as a fallback or scaling factor (`tdpi_atr_fallback`). The `tdpi_beta` coefficient from config can modify its impact based on alignment with other factors.

- **Visual Representation:**



- **Interpretation Guide:**

- High TDPI values near specific strikes suggest strong potential for price to be drawn towards or pinned at those levels due to time decay effects, especially closer to expiration.
- The shape of the TDPI curve can indicate the range of strikes most affected by immediate time decay pressure.

- **Practical Use Cases & Examples:**

- Identifying potential "pinning" strikes on expiration days.
- Assessing risk/reward for short-term options strategies sensitive to theta decay.
- The `time_decay_pin_risk` signal is directly related to TDPI.

- **Relationship to Other Metrics:** TDPI is a key component of the MSPI. It interacts with Charm/CTR (Charm Decay Rate) and TDFI (Time Decay Flow Imbalance) for more specific time-decay related signals.

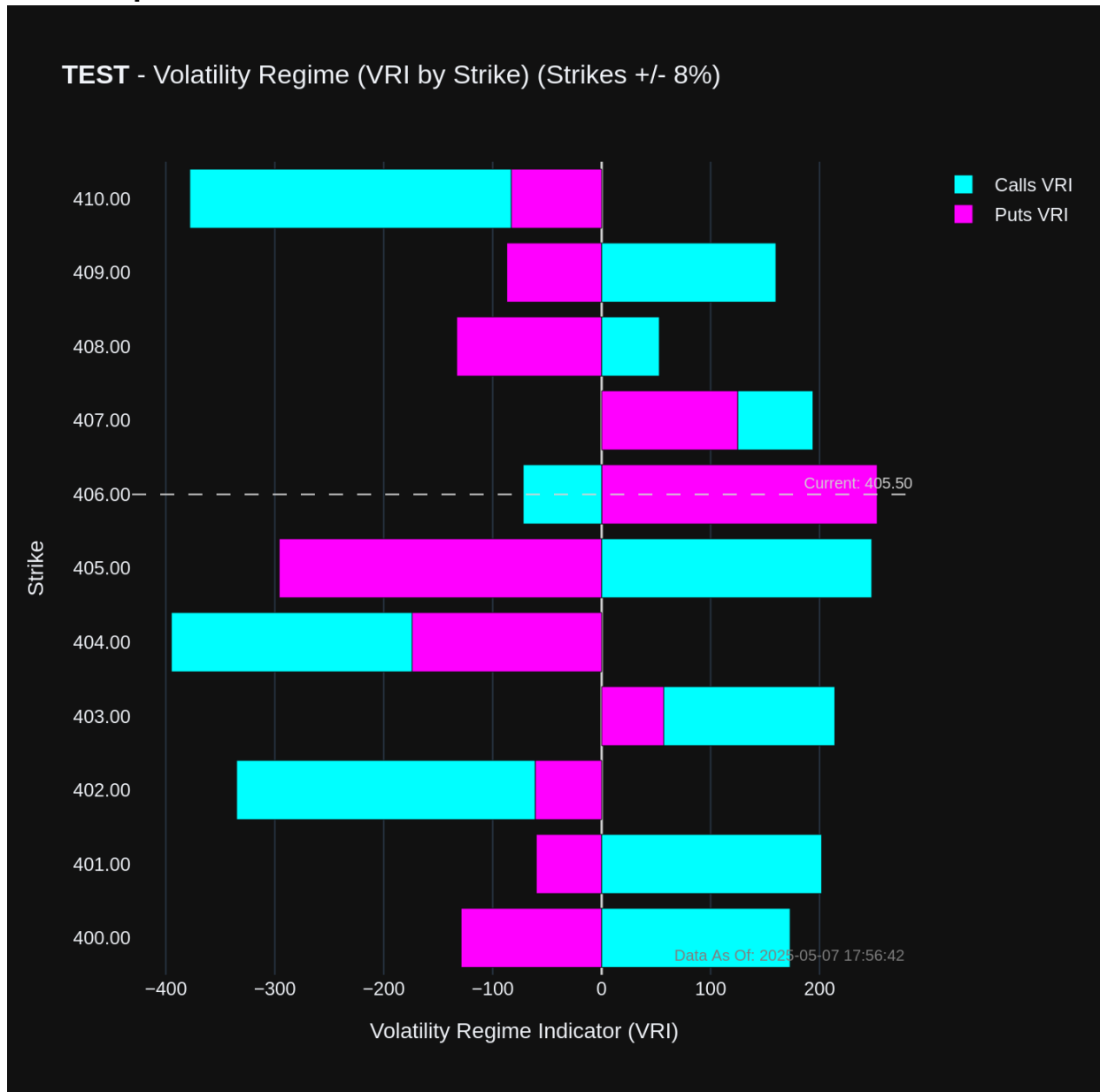
- **Configuration Notes:**

- `data_processor_settings.weights.*.tdpi` : Weight of TDPI in MSPI.
- `data_processor_settings.coefficients.tdpi_beta` : Modifies TDPI based on alignment.
- `data_processor_settings.factors.tdpi_gaussian_width` : Controls the focus of the TDPI calculation around the current price.
- `data_processor_settings.approximations.tdpi_atr_fallback` : Configuration for ATR calculation used within TDPI.
- `strategy_settings.thresholds.pin_risk_tdpi_trigger` : Threshold for generating a pin risk signal based on TDPI.

4. Volatility Risk Indicator (VRI)

- **Metric Name & Abbreviation:** Volatility Risk Indicator (VRI)
- **Conceptual Explanation:** VRI aims to quantify risks and opportunities arising from changes in implied volatility and actual price volatility. It looks at the current volatility environment (e.g., IV rank/percentile, historical volatility trends) and the market's sensitivity to volatility changes (Vega exposure).
- **Simplified Calculation Insight:** VRI is calculated in `_calculate_vri`. It likely considers Vega exposure (`vxoi`), changes in implied volatility (IV), and potentially compares historical to implied volatility. It might use a `vri_vol_trend_fallback_factor` for smoothing or when direct vol trend data is missing. The `vri_gamma` coefficient (note: this is a config name, not the Greek gamma) modifies its impact.

- **Visual Representation:**



- **Interpretation Guide:**

- High VRI might indicate a market environment prone to sharp moves if volatility expands, or opportunities if vol is expected to contract.
- Low VRI could suggest a more stable, range-bound environment from a volatility perspective.
- The `volatility_expansion` and `volatility_contraction` signals are closely tied to VRI and VFI.

- **Practical Use Cases & Examples:**

- Assessing whether to employ strategies that benefit from rising volatility (e.g., straddles) or falling volatility (e.g., credit spreads).
- Identifying periods where existing positions are at higher risk due to potential vol shifts.

- **Relationship to Other Metrics:** VRI is a component of MSPI. It works in conjunction with VFI (Volatility Flow Imbalance) to generate volatility-related trading signals.

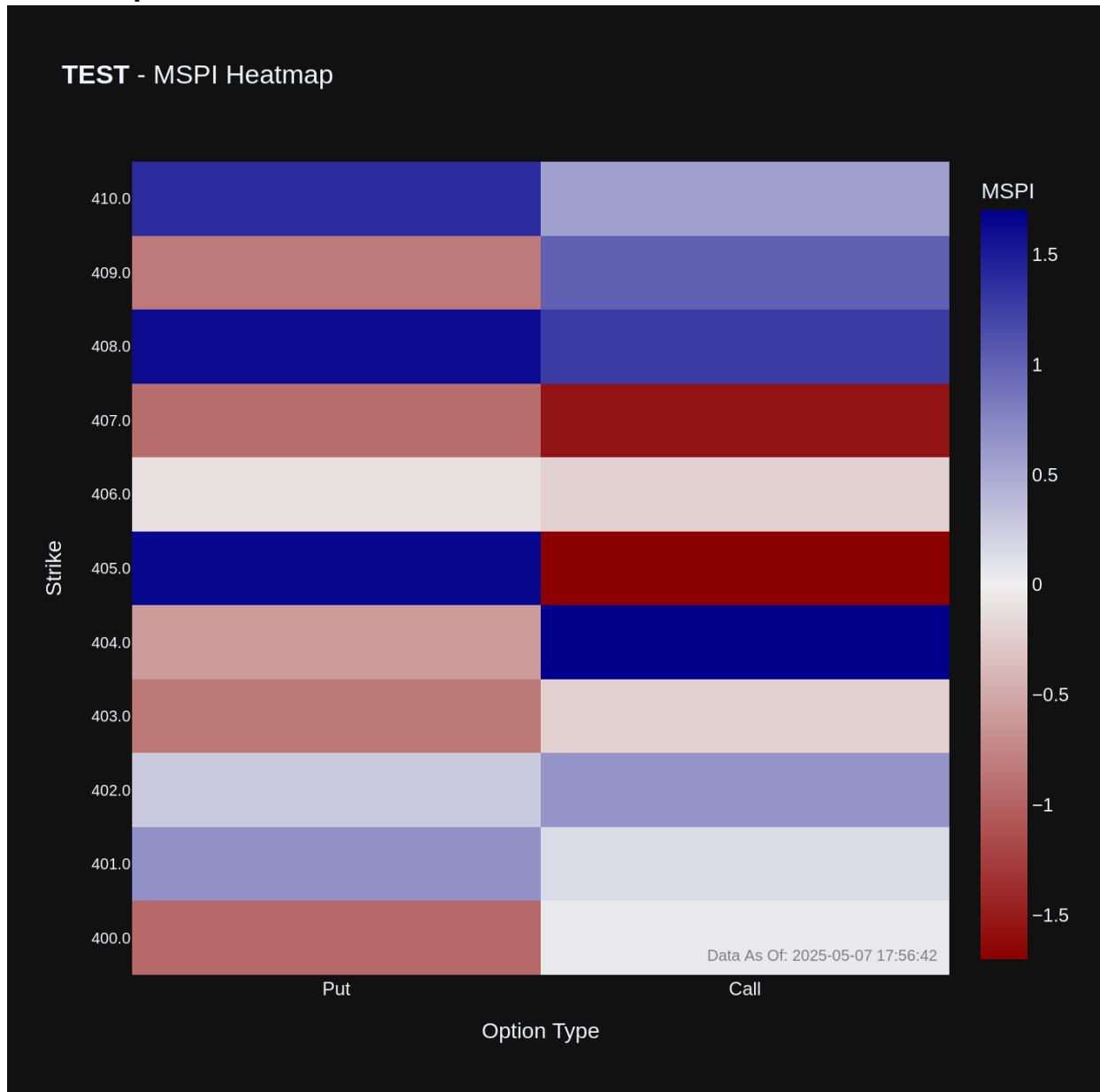
- **Configuration Notes:**

- `data_processor_settings.weights.*.*.vri` : Weight of VRI in MSPI.
- `data_processor_settings.coefficients.vri_gamma` : Modifies VRI based on alignment.
- `data_processor_settings.factors.vri_vol_trend_fallback_factor` : Fallback factor for volatility trend.
- `strategy_settings.thresholds.vol_expansion_vri_trigger` ,
`strategy_settings.thresholds.vol_contraction_vri_trigger` : Thresholds for VRI to contribute to volatility expansion/contraction signals.

5. Market Structure Position Indicator (MSPI)

- **Metric Name & Abbreviation:** Market Structure Position Indicator (MSPI)
- **Conceptual Explanation:** MSPI is the primary composite indicator of the system. It synthesizes the custom DAG, TDPI, VRI, and any enabled and weighted SDAG methodologies into a single, overarching measure of market structure pressure. The goal is to provide a holistic view of potential support/resistance and directional bias based on dealer positioning and market dynamics.
- **Simplified Calculation Insight:** MSPI is calculated in `_calculate_mspi` . It takes the normalized values of its components (DAG, TDPI, VRI, and enabled SDAGs like `sdag_multiplicative_norm` , `sdag_directional_norm` , etc.) and applies weights to them. These weights can be dynamic, based on `selection_logic` in the config (e.g., `time_based` - morning, midday, final; or `volatility_based` - low_iv, high_iv).

- **Visual Representation:**



- **Interpretation Guide:**

- High positive MSPI values at certain strikes suggest strong potential support or areas where price might be attracted upwards.
- High negative MSPI values (or low positive, depending on scaling) suggest strong potential resistance or areas where price might be pushed downwards.
- The overall MSPI profile across strikes indicates the current market structure landscape as interpreted by the system.
- Crossovers of MSPI levels or divergences with price can be significant.

- **Practical Use Cases & Examples:**

- Core indicator for identifying overall market bias and key structural levels.
- Used in conjunction with individual metrics and signals for refined trading decisions.

- **Relationship to Other Metrics:** MSPI is the aggregate of DAG, TDPI, VRI, and weighted SDAGs. Its interpretation is thus dependent on understanding its constituents.

- **Configuration Notes:**

- `data_processor_settings.weights` : This entire section dictates how MSPI components are weighted based on time or volatility context.
- All configuration notes for DAG, TDPI, VRI, and SDAGs indirectly affect MSPI as they are its inputs.

6. Sentiment Alignment Indicator (SAI)

- **Metric Name & Abbreviation:** Sentiment Alignment Indicator (SAI)
- **Conceptual Explanation:** SAI measures the degree of alignment or divergence between the MSPI (representing overall market structure) and short-term order flow dynamics (represented by OFI - Order Flow Imbalance). It helps identify if current buying/selling pressure is confirming or contradicting the underlying market structure.
- **Simplified Calculation Insight:** SAI is calculated in `_calculate_sai` . It essentially compares MSPI (or a derivative) with OFI. A positive SAI indicates alignment (e.g., bullish structure and bullish flow), while a negative SAI indicates divergence (e.g., bullish structure but bearish flow).
- **Visual Representation:** [Placeholder for SAI chart - often a histogram showing positive/negative bars, indicating alignment or divergence].
- **Interpretation Guide:**
 - Strong positive SAI: High conviction that short-term flow supports the market structure. E.g., if MSPI is bullish at a level, and OFI is also bullish, SAI will be strongly positive.
 - Strong negative SAI: High conviction that short-term flow opposes the market structure. This could signal a potential weakening of the structure or a short-term counter-trend move.
 - Values around zero: Neutral or mixed signals from flow relative to structure.
 - The `sai_high_conviction` threshold in config determines what's considered a strong signal.
- **Practical Use Cases & Examples:**
 - Confirming the strength of MSPI-derived levels. If MSPI shows support and SAI is positive, the support is considered more reliable.
 - Identifying potential reversals if SAI shows strong divergence against an established MSPI level.
- **Relationship to Other Metrics:** Directly uses MSPI and OFI (Order Flow Imbalance) as inputs.
- **Configuration Notes:**
 - `strategy_settings.thresholds.sai_high_conviction` : Defines the threshold for a high conviction SAI signal.

7. Structural Stability Index (SSI)

- **Metric Name & Abbreviation:** Structural Stability Index (SSI)
- **Conceptual Explanation:** SSI assesses the stability of the current market structure. It might look at the concentration of gamma/delta, the distance between key levels, or the rate of change in structural metrics. It helps gauge whether the market is in a stable, predictable state or a more fragile, transitional one.
- **Simplified Calculation Insight:** SSI is calculated in `_calculate_ssi`. It likely analyzes the distribution and magnitude of GEX/DEX or MSPI levels. It might incorporate concepts like the distance to the "Gamma Wall" or the density of significant strikes. It generates signals related to structure change or volatility contraction based on its value relative to configured thresholds.
- **Visual Representation:** [Placeholder for SSI chart - could be a time-series line indicating stability levels].
- **Interpretation Guide:**
 - High SSI: Generally indicates a more stable market structure, perhaps with well-defined support/resistance zones.
 - Low SSI: May suggest a less stable or transitional structure, potentially prone to breakouts or increased volatility.
 - Thresholds like `ssi_structure_change` (e.g., bottom 15th percentile) and `ssi_vol_contraction` (e.g., top 85th percentile) from config trigger specific signals.
 - `ssi_conviction_split` helps differentiate levels of conviction for SSI-based signals.
- **Practical Use Cases & Examples:**
 - Adapting trading strategy: In high SSI environments, range-trading strategies might be more effective. In low SSI, breakout or trend-following strategies might be considered.
 - Input to `complex_structure_change` and `volatility_contraction` signals.
- **Relationship to Other Metrics:** Derived from the analysis of underlying structural metrics like GEX, DEX, or MSPI.
- **Configuration Notes:**
 - `strategy_settings.thresholds.ssi_structure_change` : Threshold for detecting significant structural shifts.
 - `strategy_settings.thresholds.ssi_vol_contraction` : Threshold for identifying potential volatility contraction.
 - `strategy_settings.thresholds.ssi_conviction_split` : Threshold for conviction level within SSI signals.

8. Cumulative Flow Imbalance (CFI)

- **Metric Name & Abbreviation:** Cumulative Flow Imbalance (CFI)
- **Conceptual Explanation:** CFI tracks the net cumulative order flow over a defined period. It's similar to OFI (Order Flow Imbalance) but accumulates the values, providing a sense of sustained buying or selling pressure rather than just instantaneous flow.
- **Simplified Calculation Insight:** CFI is calculated in `_calculate_cfi`. It takes OFI values and sums them up over time. It can be reset or decay based on system logic (not explicitly detailed in `integrated_strategies_v2.py` but implied by its nature as a cumulative metric that needs a window or reset).
- **Visual Representation:** [Placeholder for CFI chart - typically a line chart showing the cumulative net flow over time].
- **Interpretation Guide:**
 - Rising CFI: Indicates sustained buying pressure.
 - Falling CFI: Indicates sustained selling pressure.
 - Divergences between CFI and price can be powerful signals (e.g., price making new highs but CFI failing to do so could indicate weakening buying momentum).
 - The `cfi_flow_divergence` threshold (with tiers) in config is used to identify significant divergences.
- **Practical Use Cases & Examples:**
 - Confirming trend strength.
 - Identifying potential trend exhaustion or reversals through divergences.
 - Input to the `complex_flow_divergence` signal.
- **Relationship to Other Metrics:** Directly derived from OFI (Order Flow Imbalance).
- **Configuration Notes:**
 - `strategy_settings.thresholds.cfi_flow_divergence` : Tiered thresholds for detecting significant flow divergences.

9. Order Flow Imbalance (OFI) - Implied

- **Metric Name & Abbreviation:** Order Flow Imbalance (OFI)
- **Conceptual Explanation:** While not explicitly calculated as a standalone output metric by `integrated_strategies_v2.py` in the same way as MSPI or SAI, OFI is a crucial input concept for SAI and CFI. It represents the net short-term buying or selling pressure derived from options order flow data (e.g., volume of buys vs. sells, or value of buys vs. sells like `volmbs` or `value_bs` from `ConvexValue` mentioned in knowledge base `user_25`, `user_28`).
- **Simplified Calculation Insight:** The system assumes OFI data is provided as part of the input DataFrame to the `process_data` method. The

`enhanced_data_processor_v2.py` would be responsible for preparing or calculating this from raw data fetched by `enhanced_data_fetcher_v2.py`. For example, it might use columns like `volm_bs` (Volume Buys - Sells) or `value_bs` (Value Buys - Sells).

- **Visual Representation:** If visualized directly, OFI would be a bar chart per strike or a time series showing net flow.
- **Interpretation Guide:**
 - Positive OFI: More buying pressure than selling pressure.
 - Negative OFI: More selling pressure than buying pressure.
 - Magnitude indicates the strength of the imbalance.
- **Practical Use Cases & Examples:**
 - Gauging immediate market sentiment.
 - Identifying strikes with unusual buying or selling activity.
- **Relationship to Other Metrics:** Input to SAI and CFI.
- **Configuration Notes:** The specific columns used for OFI would be determined by how `enhanced_data_processor_v2.py` prepares the data passed to `integrated_strategies_v2.py`. There isn't a direct config in `integrated_strategies_v2.py` for OFI itself, but rather for how metrics using OFI (like SAI, CFI) are configured.

10. Volatility Flow Imbalance (VFI) - Implied

- **Metric Name & Abbreviation:** Volatility Flow Imbalance (VFI)
- **Conceptual Explanation:** Similar to OFI, VFI is an input concept for generating volatility-related signals. It would represent the net flow into options that are sensitive to volatility changes (e.g., buying of straddles/strangles vs. selling). It helps gauge if current flow is positioned for an increase or decrease in market volatility.
- **Simplified Calculation Insight:** Assumed to be part of the input DataFrame. It might be derived from flows into specific types of options or by analyzing Vega-weighted order flow.
- **Visual Representation:** If visualized, it might show net flow into volatility-sensitive strategies.
- **Interpretation Guide:**
 - Positive VFI: Flow suggests anticipation of increasing volatility.
 - Negative VFI: Flow suggests anticipation of decreasing volatility.
- **Practical Use Cases & Examples:**
 - Anticipating shifts in implied or realized volatility based on options market positioning.

- **Relationship to Other Metrics:** Input to `volatility_expansion` and `volatility_contraction` signals, working alongside VRI.
- **Configuration Notes:**
 - `strategy_settings.thresholds.vol_expansion_vfi_trigger` , `strategy_settings.thresholds.vol_contraction_vfi_trigger` : Thresholds for VFI to contribute to volatility signals.

11. Charm Decay Rate (CTR) & Time Decay Flow Imbalance (TDFI) - Implied

- **Metric Name & Abbreviation:** Charm Decay Rate (CTR), Time Decay Flow Imbalance (TDFI)
- **Conceptual Explanation:** These are also input concepts related to time decay, specifically for the `time_decay_charm_cascade` signal.
 - **CTR** likely measures the rate of delta decay due to charm, indicating how quickly options positions will change their delta as time passes, especially near expiration.
 - **TDFI** would represent an imbalance in order flow for options highly sensitive to time decay, or perhaps a measure of net theta exposure being added/removed from the market via flow.
- **Simplified Calculation Insight:** Assumed to be part of the input DataFrame. CTR could be derived from Charm values, and TDFI from Theta-weighted flow.
- **Interpretation Guide:**
 - High CTR: Rapid delta decay, potentially leading to hedging flows.
 - TDFI imbalances: Suggests flow is positioning for or against time decay effects.
- **Practical Use Cases & Examples:**
 - Identifying risk of

IV. Trading Signals Explained

This section details the discrete trading signals generated by the `IntegratedTradingSystem` . These signals are designed to alert you to specific market conditions or potential opportunities based on the system's analysis of its underlying metrics. All signals can be toggled on or off via `config_v2.json -> system_settings -> signal_activation` .

1. Directional Signal (Bullish/Bearish/Neutral)

- **Signal Name:** Directional Signal

- **How it's Generated:** This signal is primarily derived from the **Market Structure Position Indicator (MSPI)** and confirmed or nuanced by the **Sentiment Alignment Indicator (SAI)**.
 - A **Bullish** signal typically occurs when MSPI indicates strong underlying support at current or nearby price levels, and SAI is positive (confirming order flow).
 - A **Bearish** signal typically occurs when MSPI indicates strong underlying resistance, and SAI is positive (confirming order flow in the direction of resistance) or significantly negative against support.
 - A **Neutral** signal occurs when MSPI is not showing a strong directional bias, or when MSPI and SAI are contradictory without strong conviction.
 - The exact logic is within the `_generate_trading_signals` method of `integrated_strategies_v2.py`.
- **Visual Representation:** Appears in the "Strategy Recommendations Table" on the dashboard or in the `mspi_visualizer_v2.py` output table. Example from Table:
Signal: Directional, Sentiment: Bullish, Conviction: Medium

TEST - Strategy Recommendations Table

Strike	Signal Type	Direction	Strength	Conviction	Strategy	Rationale	Target Levels	Stop Loss
405	Directional	Bullish	High	Strong	Buy Call Spread	NaN	408, 410	402
400	Volatility	Expansion	Medium	Moderate	Buy Straddle	NaN	N/A	N/A

Data As Of: 2025-05-07 17:56:42

- **Interpretation Guide:**
 - **Bullish:** Suggests the market structure and flow favor upward price movement or strong support.
 - **Bearish:** Suggests the market structure and flow favor downward price movement or strong resistance.
 - **Neutral:** Suggests a lack of clear directional pressure or conflicting signals.
 - Conviction (High/Medium/Low) is often determined by the strength of MSPI and the degree of SAI alignment.
- **Practical Use Cases & Examples:** Used as a primary indicator for taking directional trades (longs on Bullish, shorts on Bearish) or for managing existing positions.
- **Confirmation & Context:** Always use in conjunction with chart analysis of MSPI/SDAG levels. A directional signal gains strength if it occurs near a significant structural level identified by these metrics.

2. SDAG Conviction Signal (Bullish/Bearish)

- **Signal Name:** SDAG Conviction Signal
- **How it's Generated:** This signal is triggered when a configurable percentage (`sdag_conviction_threshold_pct` in `config_v2.json`) of the enabled SDAG methodologies (Multiplicative, Directional, Weighted, Volatility-Focused) agree on a sentiment (bullish or bearish) at a specific strike price. A bullish sentiment is typically indicated by SDAG values > 0 (or a positive threshold), and bearish by SDAG values < 0 (or a negative threshold).
 - (Knowledge: User_10, User_15, User_16, User_21, User_22 emphasize alignment for high conviction).
- **Visual Representation:** Appears in the "Strategy Recommendations Table". Can also be inferred by observing the alignment of SDAG methodology charts on the dashboard.
- **Interpretation Guide:**
 - **Bullish:** A significant majority of SDAG models indicate positive (supportive) pressure at a strike.
 - **Bearish:** A significant majority of SDAG models indicate negative (resistive) pressure at a strike.
 - The higher the `sdag_conviction_threshold_pct` and the more methodologies align, the stronger the conviction.
- **Practical Use Cases & Examples:** Provides high-conviction identification of support/resistance levels. Useful for pinpointing entry/exit zones or areas where price is likely to react.
- **Confirmation & Context:** Extremely powerful when these conviction strikes align with overall MSPI levels and other confirming signals.

3. Volatility Expansion Signal

- **Signal Name:** Volatility Expansion Signal
- **How it's Generated:** Triggered when the **Volatility Risk Indicator (VRI)** exceeds `vol_expansion_vri_trigger` AND the **Volatility Flow Imbalance (VFI)** (an implied input representing flow into volatility products) exceeds `vol_expansion_vfi_trigger` (both thresholds from `config_v2.json`).
- **Visual Representation:** Appears in the "Strategy Recommendations Table".
- **Interpretation Guide:** Suggests that both underlying volatility risk metrics and current market flow indicate a heightened probability of an increase in market volatility (price swings becoming larger/faster).
- **Practical Use Cases & Examples:**
 - Consider strategies that profit from rising volatility (e.g., long straddles, strangles).
 - Widen profit targets and stop-losses on existing directional trades.
 - Exercise caution, as increased volatility can lead to unpredictable moves.
- **Confirmation & Context:** Look for price coiling or consolidation on charts, which often precedes volatility expansion. Also, check if price is near key "Volatility Trigger" levels identified by SDAGs (Knowledge: User_33).

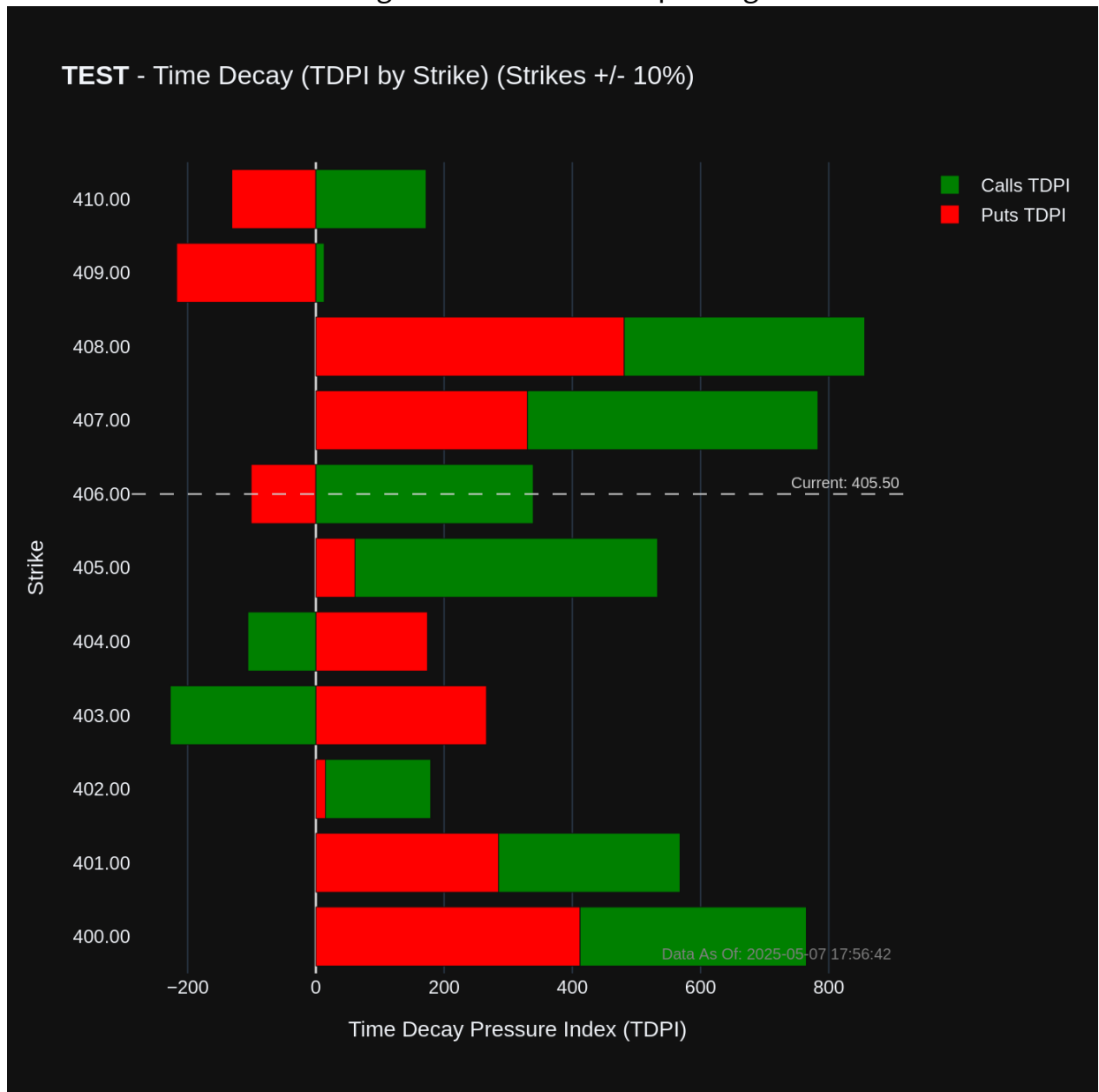
4. Volatility Contraction Signal

- **Signal Name:** Volatility Contraction Signal
- **How it's Generated:** Triggered when the **Volatility Risk Indicator (VRI)** falls below `vol_contraction_vri_trigger` AND **Volatility Flow Imbalance (VFI)** falls below `vol_contraction_vfi_trigger` AND the **Structural Stability Index (SSI)** is above its `ssi_vol_contraction` threshold (indicating a stable structure conducive to vol contraction).
- **Visual Representation:** Appears in the "Strategy Recommendations Table".
- **Interpretation Guide:** Suggests that market conditions favor a decrease in volatility, potentially leading to more range-bound or subdued price action.
- **Practical Use Cases & Examples:**
 - Consider strategies that profit from falling or stagnant volatility (e.g., short straddles/strangles, credit spreads, iron condors).
 - Expect more defined trading ranges around key MSPI/SDAG levels.
- **Confirmation & Context:** Often occurs after a period of high volatility. Confirm with SSI indicating a return to a more stable structure.

5. Time Decay Pin Risk Signal

- **Signal Name:** Time Decay Pin Risk Signal

- **How it's Generated:** Triggered when the **Time Decay Pressure Indicator (TDPI)** at or very near the current underlying price exceeds the `pin_risk_tdpi_trigger` threshold from `config_v2.json`.
- **Visual Representation:** Appears in the "Strategy Recommendations Table". The TDPI chart itself will show high values around the pinning strike.



- **Interpretation Guide:** Indicates a high probability that the underlying price will be drawn towards and potentially "pinned" to a specific strike price, especially as options expiration approaches, due to accelerating time decay (theta) and associated hedging.
- **Practical Use Cases & Examples:**
 - Useful for options selling strategies targeting specific strikes on expiration week/day (e.g., iron butterflies, short straddles at the pin strike).
 - Can also indicate areas where price movement might stall.
- **Confirmation & Context:** Most relevant for options nearing expiration. Check for high open interest at the potential pin strike.

6. Time Decay Charm Cascade Signal

- **Signal Name:** Time Decay Charm Cascade Signal
- **How it's Generated:** Triggered when the **Charm Decay Rate (CTR)** (implied input) exceeds `charm_cascade_ctr_trigger` AND the **Time Decay Flow Imbalance (TDFI)** (implied input) exceeds `charm_cascade_tdfi_trigger`. This signal relates to the rapid change in options' deltas due to charm, especially for near-expiration options, potentially forcing significant dealer hedging.
- **Visual Representation:** Appears in the "Strategy Recommendations Table".
- **Interpretation Guide:** Suggests a potential for accelerated price movement as dealers adjust hedges due to rapid delta decay caused by charm. This effect is most pronounced for options close to expiry and away from at-the-money.
- **Practical Use Cases & Examples:** Can signal late-day directional moves on or just before expiration, as positions are unwound or hedged aggressively.
- **Confirmation & Context:** Highly specific to expiration dynamics. Look for unusual volume in near-term options.

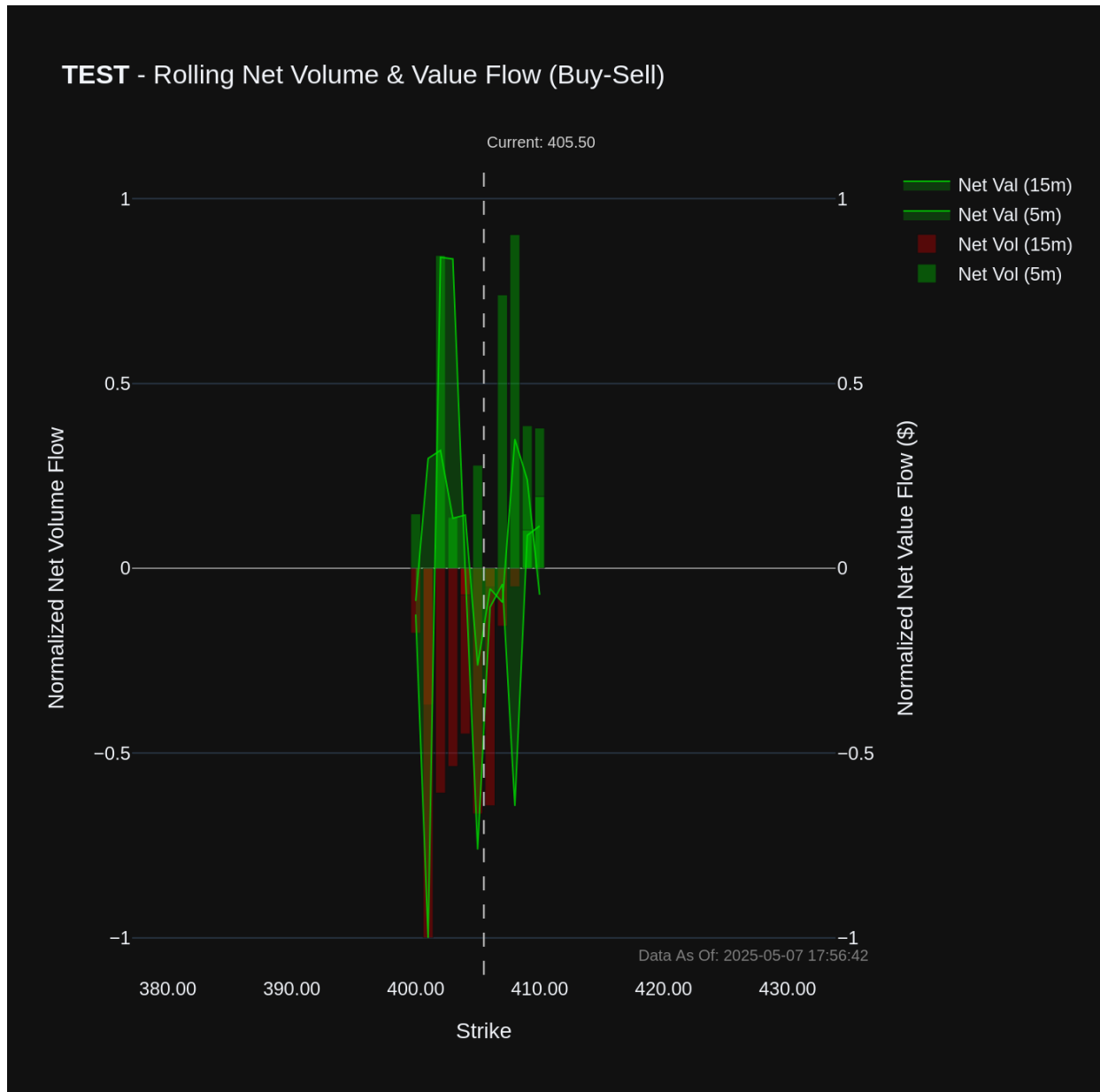
7. Complex Structure Change Signal

- **Signal Name:** Complex Structure Change Signal
- **How it's Generated:** Triggered when the **Structural Stability Index (SSI)** falls below its `ssi_structure_change` threshold (e.g., a low percentile value, indicating instability).
- **Visual Representation:** Appears in the "Strategy Recommendations Table".
- **Interpretation Guide:** Warns that the prevailing market structure (as defined by GEX/DEX/MSPI levels) may be breaking down or shifting significantly. This can precede a transition to a new trend or a period of increased uncertainty.
- **Practical Use Cases & Examples:**
 - A caution signal: existing range-bound strategies may become less reliable.
 - May signal an opportunity for breakout strategies if other indicators confirm a new direction.
- **Confirmation & Context:** Look for price challenging or breaking previously strong MSPI/SDAG levels. Low SSI often accompanies such moves.

8. Complex Flow Divergence Signal

- **Signal Name:** Complex Flow Divergence Signal
- **How it's Generated:** Triggered when the **Cumulative Flow Imbalance (CFI)** shows a significant divergence from price action, exceeding the `cfi_flow_divergence`

tiered thresholds.



- **Interpretation Guide:**
 - **Bullish Divergence:** Price makes a new low, but CFI makes a higher low (selling pressure is waning). Suggests potential upward reversal.
 - **Bearish Divergence:** Price makes a new high, but CFI makes a lower high (buying pressure is waning). Suggests potential downward reversal.
 - The tiered thresholds in config allow for different levels of divergence sensitivity.
- **Practical Use Cases & Examples:** Powerful for identifying potential trend exhaustion and spotting early signs of reversals.
- **Confirmation & Context:** Most effective when observed at key support/resistance levels or after extended trends.

V. Cohesive Analysis: Integrating Metrics & Signals

Understanding individual metrics and signals is the foundation, but the true power of this trading system comes from synthesizing this information into a cohesive market view. This section focuses on how to combine various components to make more informed trading decisions.

1. The Big Picture: How Individual Components Work Together

The system is designed around a core idea: market maker hedging and significant order flow can create predictable (or at least probabilistically favorable) patterns and levels in the market.

- **Structural Foundation (MSPI & its components - DAG, SDAGs, TDPI, VRI):** These metrics establish the underlying market structure. High MSPI/SDAG values at specific strikes suggest strong support/resistance due to concentrated dealer positions. TDPI highlights time-decay pressures, especially near expiration, while VRI assesses the volatility environment.
- **Flow Confirmation/Divergence (SAI, CFI, OFI):** These metrics overlay short-term and cumulative order flow onto the established structure. Positive SAI (alignment) strengthens the conviction in MSPI levels, while negative SAI (divergence) can signal caution or potential turning points. CFI tracks sustained pressure.
- **Stability Assessment (SSI):** SSI tells you how robust the current structure is. A stable structure (high SSI) might favor range-bound strategies around key MSPI/SDAG levels. A less stable structure (low SSI) might suggest caution or preparedness for breaks.
- **Specific Event Triggers (Volatility Signals, Pin Risk, Charm Cascade, Structure Change, Flow Divergence):** These are discrete signals generated when specific thresholds (defined in `config_v2.json`) are met by the underlying metrics. They act as alerts for potentially actionable conditions.

The Goal: To find confluence. When multiple indicators point in the same direction, the conviction level for a trading idea increases. For example: - Strong MSPI/SDAG support level + Positive SAI + High SSI + Bullish directional signal = Higher conviction long entry. - Strong MSPI/SDAG resistance level + Negative SAI (flow diverging) + Low SSI + Bearish signal = Potential short opportunity or caution for longs.

2. Workflow Examples

Let's walk through a few hypothetical scenarios to illustrate how to use the system. (Note: These examples will refer to visual placeholders. In a real guide, these would be actual chart images from the dashboard or visualizer.)

Scenario 1: Identifying a High-Conviction Bullish Setup

1. **Initial Scan (Dashboard/Visualizer):** You observe the MSPI chart. Price is approaching a significant strike level where MSPI shows a strong positive peak. Multiple SDAG methodologies (Multiplicative, Directional, Weighted) also show strong positive peaks (>1.0 , ideally >1.5) at or very near this strike. (Knowledge: User_10, User_15, User_19, User_21)
 - [Visual Placeholder: MSPI chart showing price near a strong positive MSPI/SDAG cluster]
2. **Check Order Flow (SAI, OFI):** The Sentiment Alignment Indicator (SAI) for this strike zone is strongly positive, indicating that recent order flow is bullish and supports the structural strength.
 - [Visual Placeholder: SAI histogram showing positive bars for the relevant price area]
3. **Assess Stability (SSI):** The Structural Stability Index (SSI) is in a moderate to high range, suggesting the identified support level is relatively stable and not immediately prone to collapse.
4. **Look for Specific Signals:** The dashboard might show a "Directional Bullish" signal or a "SDAG Conviction Bullish" signal if the criteria are met (e.g., `sdag_conviction_threshold_pct` of SDAGs agree).
5. **Consider Volatility (VRI, Volatility Signals):** VRI is neutral or low, and no "Volatility Expansion Bearish" signals are present. This means a sudden volatility spike against the position is less likely.
6. **Decision Point:** The confluence of strong MSPI/SDAG support, confirming bullish order flow (SAI), reasonable structural stability (SSI), and potentially a discrete bullish signal suggests a high-probability long entry near the identified strike. Stop-loss could be placed below this structural level.

Scenario 2: Assessing Choppy or Range-Bound Market Conditions

1. **Initial Scan:** MSPI and SDAG charts show multiple significant positive and negative levels relatively close to each other, effectively bracketing the current price. There isn't a clear, dominant single direction from the overall profile. The "Wall" level (largest GEX/DAG concentration) might be clearly defined. (Knowledge: User_35, User_38, User_43)
 - [Visual Placeholder: MSPI/SDAG chart showing price oscillating between defined positive and negative zones]
2. **Order Flow:** SAI might be flipping between positive and negative without strong conviction, or OFI might show balanced flow.
3. **Stability:** SSI could be moderate, indicating the range is somewhat established.

4. **Signals:** You might see alternating short-term directional signals that don't lead to sustained moves, or an absence of strong conviction signals.
5. **Decision Point:** This scenario suggests a range-trading environment. Trades could be initiated near the boundaries of the MSPI/SDAG-defined range, targeting the other side. Focus on levels where multiple SDAG methodologies align. (Knowledge: User_10, User_43). Profit targets should be modest, and risk management tight, as true breakouts are not expected.

Scenario 3: Detecting Potential Volatility Expansion

1. **Initial Scan (VRI, SDAGs):** VRI starts to trend higher, or you notice SDAG values becoming extremely negative (e.g., < -1.5) at certain strikes, indicating potential "Volatility Trigger" zones. (Knowledge: User_15, User_19, User_33)
 - [Visual Placeholder: VRI chart trending up, or SDAG chart showing a significant negative trough below -1.5]
2. **Check SSI:** SSI might be low or decreasing, suggesting the market structure is becoming less stable and more prone to a sharp move.
3. **Signals:** A "Volatility Expansion" signal might trigger on the dashboard, based on VRI and VFI thresholds.
4. **Price Action:** Price might be coiling or consolidating, often a precursor to a volatility event.
5. **Decision Point:** This setup warns of a potential sharp increase in volatility. Strategies that benefit from volatility expansion (e.g., long straddles, strangles) could be considered. Alternatively, existing positions might need tighter risk management or hedging. If price breaks a key Volatility Trigger level, it can lead to accelerated movement. (Knowledge: User_33)

3. Developing a Trading Plan: Using the System to Form Hypotheses

A robust trading plan involves more than just reacting to signals. Use the system to:

- **Form a Market Thesis:** Based on the overall MSPI/SDAG profile, SSI, and CFI, what is the dominant market condition? (Trending, ranging, volatile, quiet?)
- **Identify Key Levels:** Mark the most significant support/resistance levels identified by MSPI and SDAG alignment. Note any "Wall" levels or "Volatility Triggers".
- **Define Entry Criteria:** What specific confluence of metrics and signals will trigger an entry? (e.g., Price at MSPI support + Positive SAI + Bullish SDAG Conviction Signal).
- **Set Stop-Loss Points:** Typically based on structural levels. If a key support level is breached decisively, the thesis might be invalidated.
- **Determine Profit Targets:** Other significant MSPI/SDAG levels, or based on risk/reward ratios.

- **Adapt to Changing Conditions:** The market is dynamic. Continuously monitor the dashboard and be prepared to adjust your thesis and plan as new data comes in. (Knowledge: User_15, User_21 - SDAG values are dynamic).

4. Customizing Your View: Leveraging Dashboard Features and `config_v2.json`

The system is designed to be adaptable:

- **Dashboard Interactivity:** Use the dashboard's controls (symbol, DTE, range selection, refresh interval) to focus on the specific market segments and timeframes relevant to your trading style.
- **`config_v2.json` Deep Dive:** For advanced users, `config_v2.json` offers extensive customization:
 - **Weights (`data_processor_settings.weights`):** Adjust how MSPI components (DAG, TDPI, VRI, SDAGs) are weighted. You can favor certain metrics based on your observations or market regime (e.g., increase TDPI weight closer to expiration).
 - **Thresholds (`strategy_settings.thresholds`):** Fine-tune the sensitivity of various signals (SAI, SSI, CFI, Volatility, Pin Risk, etc.). Lowering a threshold makes a signal trigger more easily; raising it requires stronger conditions.
 - **SDAG Methodologies (`strategy_settings.dag_methodologies`):** Enable/disable specific SDAG calculations, adjust their individual weights, and their contribution to the overall MSPI and conviction signals.
 - **Visualization Settings:** Customize colors, chart types, and default parameters for visual outputs to suit your preferences.

By understanding these customization options, you can tailor the system's output to better align with your unique trading strategies and market interpretations.

VI. Visual Guide to the Dashboard & Charts

This section provides an overview of the `enhanced_dashboard_v2` and the charts generated by `mspi_visualizer_v2.py`, helping you navigate and interpret the visual information provided by the system.

1. Overview of the `enhanced_dashboard_v2` Layout

The `enhanced_dashboard_v2` is designed for a clear and interactive experience. (As this is a text-based guide, detailed layout descriptions will be provided. In a live system, screenshots would be here.)

Typically, the dashboard is structured with:

- **Control Panel:** Usually at the top or side, allowing you to select:
 - Underlying Symbol (e.g., SPY, AAPL)
 - Days to Expiration (DTE)
 - Price Range Percentage (for focusing charts around current price)
 - Data Refresh Interval
 - Potentially toggles for specific chart views or overlays.
- **Main Display Area:** This area houses the various charts and tables, organized into logical sections or tabs.
 - **Key Metrics Overview:** Charts for MSPI, individual SDAG methodologies, potentially TDPI and VRI profiles across strikes.
 - **Flow Analysis:** Charts for SAI, CFI, and potentially raw OFI visualizations.
 - **Signal Summary:** A table or display listing currently active trading signals.
 - **Volatility Insights:** Charts related to VRI, VFI, and volatility trigger levels.
 - **Detailed Greek Profiles:** Visualizations of raw Gamma, Delta, Vega, Theta across strikes.
- **Status Bar/Log:** Displays system status, last update time, and any important messages or errors.

2. Detailed Explanation of Key Charts (Examples from `mspi_visualizer_v2.py`)

The `mspi_visualizer_v2.py` script can generate several standalone charts that are also representative of what you might see on the dashboard. We will use the PNG files generated in `/home/ubuntu/mspi_visualizer_v2_test_charts/` as references.

- **MSPI Heatmap (`test_mspi_heatmap_*.png`)**
 - **What it shows:** This chart typically displays MSPI values (or a similar composite structural metric) across a range of strike prices (Y-axis) over time (X-axis). Colors indicate the strength and polarity of MSPI, with warmer colors (e.g., reds) for positive/supportive MSPI and cooler colors (e.g., blues) for negative/resistive MSPI, or vice-versa depending on the chosen `color_scales` in `config_v2.json` .
 - **How to interpret:** Look for persistent horizontal bands of strong color, indicating stable structural levels. Changes in color intensity or shifts in these bands can signal evolving market structure. The current price interaction with these colored zones is key.
 - [Visual Placeholder: Embed `test_mspi_heatmap_*.png` here with annotations]

- **Net Value Pressure Heatmap (`test_net_value_pressure_heatmap_*.png`)**

- **What it shows:** Similar to the MSPI heatmap, but this likely visualizes a metric based on net dollar value of options (e.g., Delta value or Gamma value) across strikes and time. It focuses on where the most significant financial exposure is concentrated.
- **How to interpret:** Identifies levels with large financial weight, which can act as magnets or areas of significant hedging activity. Useful for seeing where large positions might influence price.
- [Visual Placeholder: Embed `test_net_value_pressure_heatmap_*.png` here with annotations]

- **Raw Time Decay Chart (TDPI by Strike) (`test_raw_time_decay_chart_*.png`)**

- **What it shows:** This chart displays the Time Decay Pressure Indicator (TDPI) values across different strike prices for a specific point in time. It usually peaks around the current underlying price.
- **How to interpret:** Higher TDPI values indicate strikes where time decay pressure is most acute. This is particularly relevant for identifying potential pinning strikes as expiration approaches. The width of the peak shows the range of influence.
- [Visual Placeholder: Embed `test_raw_time_decay_chart_*.png` here with annotations]

- **Raw Volatility Regime Chart (VRI by Strike)
(`test_raw_volatility_regime_chart_*.png`)**

- **What it shows:** This visualizes the Volatility Risk Indicator (VRI) across strikes. It helps identify strikes that are more sensitive to changes in volatility or where current volatility conditions pose a greater risk/opportunity.
- **How to interpret:** Peaks in VRI might indicate strikes where Vega exposure is high or where the market perceives higher volatility risk. Useful for structuring volatility-based trades.
- [Visual Placeholder: Embed `test_raw_volatility_regime_chart_*.png` here with annotations]

- **Combined Rolling Flow Chart (`test_combined_rolling_flow_*.png`)**

- **What it shows:** This chart likely displays rolling sums or averages of order flow metrics like Net Volume (Buy-Sell Volume) and Net Value (Buy-Sell Value) over time. This is a way to visualize CFI or similar cumulative flow metrics.
- **How to interpret:** Rising lines indicate sustained buying pressure in terms of volume or value; falling lines indicate selling pressure. Divergences between

these flow lines and the price can be powerful leading indicators for potential trend changes or exhaustion.

- [Visual Placeholder: Embed `test_combined_rolling_flow_*.png` here with annotations]

- **Strategy Recommendations Table**

(`test_strategy_recommendations_table_*.png`)

- **What it shows:** This is a tabular representation of the trading signals generated by the `IntegratedTradingSystem` . It lists active signals, their direction (bullish/bearish/neutral), conviction level (if applicable), and potentially the metrics that triggered them.
- **How to interpret:** Provides a quick summary of actionable alerts from the system. Always cross-reference with the underlying charts to understand the context of the signal.
- [Visual Placeholder: Embed `test_strategy_recommendations_table_*.png` here with annotations]

Other Potential Dashboard Charts (Conceptual based on system capabilities):

- **Strike-Based SDAG Comparison:** A chart showing lines for each enabled SDAG methodology (Multiplicative, Directional, etc.) plotted against strike prices. Excellent for identifying consensus levels.
- **SAI Histogram:** Bars per strike showing positive (flow aligning with structure) or negative (flow diverging from structure) SAI values.
- **CFI Line Chart:** A time series of the Cumulative Flow Imbalance.
- **SSI Time Series:** A line chart showing the Structural Stability Index over time.

3. Key Interactive Features

Dashboards built with Plotly/Dash typically offer:

- **Tooltips:** Hovering over data points on charts reveals detailed information (e.g., exact metric value, strike price, date/time).
- **Zoom & Pan:** Ability to zoom into specific areas of a chart for closer inspection and pan across the data.
- **Clickable Legends:** Toggle the visibility of different traces on a chart by clicking their names in the legend.
- **Cross-filtering (Potentially):** Selecting a data point or range on one chart might filter or highlight corresponding data on other charts (if implemented).

VII. Advanced Configuration & Customization

This section provides a deeper look into `config_v2.json`, enabling advanced users to tailor the system's behavior to their specific needs and market views. Always back up your `config_v2.json` before making significant changes.

1. Deep Dive into `config_v2.json` Sections

(Refer to your `config_v2.json` and `config.schema.json` for the full structure. Key areas for trading logic customization are highlighted below.)

- **`data_processor_settings.weights`** :
 - `selection_logic` : Choose between "time_based" (weights change for morning, midday, final hours) or "volatility_based" (weights change based on IV percentile).
 - `time_based` & `volatility_based` sub-sections: Define the actual percentage weights for `dag_custom`, `tdpi`, `vri`, and each enabled `sdag_[method_name]_norm` that contributes to MSPI. The sum of these weights for any given period/context should ideally be close to 1.0.
 - Impact: Directly alters the composition and sensitivity of the MSPI. If you believe TDPI is more critical mid-day, you can increase its weight for that period.
- **`data_processor_settings.coefficients`** : (e.g., `dag_alpha`, `tdpi_beta`, `vri_gamma`)
 - These apply multipliers to the respective metrics based on whether their internal components are "aligned", "opposed", or "neutral".
 - Impact: Allows you to amplify or dampen the effect of a metric when its underlying drivers are strongly confirming or conflicting. For example, a `dag_alpha.aligned` of 1.3 boosts DAG when its gamma and delta components suggest similar hedging pressure.
- **`data_processor_settings.factors`** : (e.g., `tdpi_gaussian_width`, `vri_vol_trend_fallback_factor`)
 - Fine-tunes specific aspects of metric calculations.
 - `tdpi_gaussian_width` : A negative value (e.g., -0.5) typically makes the Gaussian curve wider, meaning TDPI considers strikes further from the current price. A value closer to zero would make it narrower, focusing pressure more tightly.

- Impact: Modifies the shape and reach of individual metric calculations.
- **strategy_settings.gamma_exposure_source_col , delta_exposure_source_col , skew_adjusted_gamma_source_col :**
 - Defines which columns from your input data are used for GEX, DEX, and Skew-Adjusted GEX.
 - Impact: Critical for ensuring the system uses the correct data. `gxoi` (gamma * open interest) and `dxoi` (delta * open interest) are common. If you have skew-adjusted GEX data, specify its column name and set `use_skew_adjusted_for_sdag` to true.
- **strategy_settings.use_skew_adjusted_for_sdag :**
 - Boolean (true/false). If true, SDAG calculations will use the `skew_adjusted_gamma_source_col` . If false, they use the standard `gamma_exposure_source_col` .
 - Impact: Determines if SDAG methodologies incorporate volatility skew, which can provide a more realistic view of hedging pressures (Knowledge: User_6, User_8).
- **strategy_settings.dag_methodologies :**
 - `enabled` : A list of SDAG methodologies to calculate (e.g., ["multiplicative", "directional"]).
 - For each methodology (e.g., `multiplicative`):
 - `delta_weight_factor` (or `w1_gamma` , `w2_delta` for weighted): Parameters specific to that SDAG formula.
 - `weight_in_mspi` : How much this normalized SDAG methodology contributes to the overall MSPI score. If 0, it's calculated for analysis/signals but not part of MSPI.
 - Impact: Controls which advanced SDAG views are available and how they influence the composite MSPI.
- **strategy_settings.sdag_conviction_threshold_pct :**
 - Percentage (0.0 to 1.0). The proportion of enabled SDAG methodologies that must agree on a sentiment (bullish/bearish at a strike) to generate an `sdag_conviction` signal.
 - Impact: A higher value (e.g., 0.75 meaning 3 out of 4 enabled must agree) leads to fewer but higher-conviction SDAG agreement signals.

- **strategy_settings.thresholds :**

- This is a large section defining trigger points for various signals (SAI, SSI, CFI, Volatility Expansion/Contraction, Pin Risk, Charm Cascade).
- Each threshold has a `type` (`fixed` , `relative_percentile` , `relative_mean_factor`) and a `value` (or `percentile` , `factor`). Many also have a `fallback_value` .
- Impact: Directly controls the sensitivity of all discrete trading signals. Adjusting these requires careful consideration and testing, as it changes how readily the system will alert you to specific conditions.
 - Example: `sai_high_conviction.value` : If 0.7, SAI must be >0.7 or <-0.7 for a high conviction signal.
 - Example: `ssi_structure_change.percentile` : If 15, an SSI value in the bottom 15th percentile of its recent range will trigger a structure change warning.

2. How to Adjust Parameters for Different Market Conditions or Risk Appetites

- **Volatile Markets:**

- Consider increasing the weight of VRI in MSPI.
- Lower thresholds for Volatility Expansion signals to be alerted sooner.
- Potentially rely more on shorter-term SDAG methodologies (Multiplicative, Directional).

- **Quiet/Ranging Markets:**

- Increase weight of TDPI if near expiration.
- Focus on SDAG levels for range boundaries.
- Adjust thresholds for CFI/SAI to catch divergences that might signal a range break.

- **Higher Risk Appetite:**

- May lower conviction thresholds slightly to get more signals, understanding this may increase false positives.

- **Lower Risk Appetite:**

- Increase conviction thresholds (e.g., `sai_high_conviction` , `sdag_conviction_threshold_pct`).
- Require stronger alignment across multiple metrics before acting.

3. Understanding the Impact of Configuration Changes

- **Test Incrementally:** Change one group of related parameters at a time and observe the impact on historical data or a paper trading account before applying to live trading.
- **Normalization is Key:** Remember that MSPI components (DAG, TDPI, VRI, SDAGs) are normalized before being weighted. Changing the calculation of one component (e.g., via its factors or coefficients) can affect its normalized range and thus its influence on MSPI, even if its weight remains the same.
- **Documentation is Crucial:** Keep a personal log of changes made to `config_v2.json` and the observed outcomes. This will help you refine your configuration over time.

By mastering the configuration, you transform the system from a generic tool into a personalized trading assistant that reflects your market understanding and risk preferences.

VI. Visual Guide: Navigating the Dashboard & Charts

This section is intended to provide a detailed walkthrough of the `enhanced_dashboard_v2.py` application and all charts generated by the system, including those from `mspi_visualizer_v2.py`.

(Placeholder: This section will be populated with screenshots and explanations of the dashboard layout, individual chart interpretations (MSPI/SDAG levels, Flow Charts, Volatility Charts, Time Decay Charts, Strategy Recommendation Table), and how to interact with dashboard elements. For now, please refer to the generated images in the `images` folder accompanying this guide for examples of the visual outputs.)

Key visualizations include: - **MSPI Heatmap:** Shows the Market Structure Position Indicator across different strikes and potentially timeframes, highlighting areas of strong structural support or resistance. (Example: `images/test_mspi_heatmap_20250507_175642.png`) - **Combined Rolling Flow (CFI):** Tracks cumulative flow imbalances, useful for spotting divergences. (Example: `images/test_combined_rolling_flow_20250507_175645.png`) - **Raw Time Decay Chart (TDPI):** Illustrates time decay pressure around strike prices. (Example: `images/test_raw_time_decay_chart_20250507_175645.png`) - **Raw Volatility Regime Chart (VRI-related):** Provides insights into the current volatility environment. (Example: `images/test_raw_volatility_regime_chart_20250507_175645.png`) - **Net Value Pressure Heatmap:** Visualizes net value pressure from options flow. (Example: `images/test_net_value_pressure_heatmap_20250507_175645.png`) - **Strategy**

Recommendations Table: Summarizes key signals and actionable recommendations.
(Example: `images/test_strategy_recommendations_table_20250507_175646.png`)

You can run `python3.11 /home/ubuntu/mspi_visualizer_v2.py` to generate these sample charts. The dashboard (`python3.11 -m dashboard_v2.enhanced_dashboard_v2`) provides an interactive way to explore this data.

VII. Advanced Configuration & Customization

This section is intended to offer a deep dive into `config_v2.json` for fine-tuning the system's behavior.

(Placeholder: This section will detail how to adjust thresholds, weights for metrics in MSPI, enable/disable specific metrics or signals, and understand the impact of `selection_logic` (time_based, volatility_based) on the system's output. For now, please refer to the `config_v2.json` file itself and the comments within `integrated_strategies_v2.py` and other relevant scripts for insights into configuration options.)

Key areas for customization in `config_v2.json` include: - `strategy_settings` : Thresholds for signals, SDAG methodology parameters, source column definitions. - `data_processor_settings` : Weights for MSPI components, coefficients, factors, and approximation settings. - `system_settings` : Signal activation toggles, logging levels. - `visualization_settings` : Dashboard refresh rates and chart-specific settings.

VIII. Troubleshooting & FAQ

- **Q1: The dashboard is not loading data or shows errors.**
 - **A1:**
 - Check `config_v2.json` : Ensure `api_credentials` (email, password, environment) for ConvexValue are correctly entered if you intend to fetch live data. If not, the system will use dummy data for the dashboard if `enhanced_data_fetcher_v2.py` is set up to do so.
 - Ensure `enhanced_data_fetcher_v2.py` and `enhanced_data_processor_v2.py` are running correctly and producing output in the `data_directory` specified in `config_v2.json` if the dashboard reads from files.
 - Check the console output where you launched `python3.11 -m dashboard_v2.enhanced_dashboard_v2` for any specific error messages.

- Verify that `convexlib` is installed correctly (`pip3 install git+https://github.com/convexvalue/convexlib.git`).
- **Q2: A specific metric (e.g., SDAG) is not showing up or seems incorrect.**
 - **A2:**
 - Check `config_v2.json` -> `strategy_settings` -> `dag_methodologies` : Ensure the SDAG methodology you are interested in is listed in `enabled` .
 - Verify the `gamma_exposure_source_col` , `delta_exposure_source_col` , and `skew_adjusted_gamma_source_col` (if `use_skew_adjusted_for_sdag` is true) are correctly mapped to the columns in your input data.
 - Examine the log files (if detailed logging is enabled via `system_settings.log_level`) for any warnings or errors during the calculation of that metric by `integrated_strategies_v2.py` or `enhanced_data_processor_v2.py` .
- **Q3: Signals seem too frequent or too rare.**
 - **A3:** This is likely related to the threshold settings in `config_v2.json` -> `strategy_settings` -> `thresholds` .
 - If signals are too frequent, consider increasing the `value` for fixed thresholds, or using a more extreme `percentile` or `factor` for relative thresholds.
 - If signals are too rare, consider decreasing these values. Make changes incrementally and observe the impact.
- **Q4: How do I interpret a very high or very low MSPI/SDAG value?**
 - **A4:** Generally, for MSPI/SDAG, values significantly above zero (e.g., > 1.0 or 1.5 as per knowledge base User_19) indicate strong positive structural influence (support/magnet). Values significantly below zero (e.g., < -1.0 or -1.5) indicate strong negative influence (resistance/volatility trigger). The exact interpretation depends on the specific metric and its normalization. Refer to Section III for details on each metric.
- **Q5: The system uses `gxoi` and `dxoi` . What if my data provider uses different names?**
 - **A5:** You can change the source column names in `config_v2.json` -> `strategy_settings` (e.g., `gamma_exposure_source_col` ,

`delta_exposure_source_col`). Ensure your data processing pipeline provides these columns to the `integrated_strategies_v2.py` module.

- **Q6: How can I improve the performance of the dashboard or data processing?**
 - **A6:**
 - For the dashboard, increase the `refresh_interval_ms` in `config_v2.json` -> `visualization_settings` -> `dashboard` -> `defaults` to reduce the frequency of data updates if real-time updates are not critical.
 - Ensure your machine has sufficient resources (CPU, RAM).
 - For data processing, the efficiency largely depends on the Pandas operations within the scripts. The V2 scripts have been optimized, but very large datasets will always take time.

IX. Glossary of All Metrics & Signals

- **CFI (Cumulative Flow Imbalance):** Tracks net cumulative order flow over time.
- **CTR (Charm Decay Rate):** Measures the rate of delta decay due to charm (time passage effect on delta).
- **DAG (Delta Adjusted Gamma Exposure - Custom):** Proprietary metric combining gamma and delta exposures to identify key structural levels.
- **DEX (Delta Exposure):** Net delta sensitivity of options positions.
- **GEX (Gamma Exposure):** Net gamma sensitivity of options positions.
- **MSPI (Market Structure Position Indicator):** Primary composite indicator synthesizing DAG, TDPI, VRI, and SDAGs.
- **OFI (Order Flow Imbalance):** Net short-term buying or selling pressure from options order flow.
- **SAI (Sentiment Alignment Indicator):** Measures alignment between MSPI (structure) and OFI (flow).
- **SDAG (Skew and Delta Adjusted GEX):** Advanced metric combining (skew-adjusted) GEX with DEX. Multiple methodologies exist (Multiplicative, Directional, Weighted, Volatility-Focused).
- **SSI (Structural Stability Index):** Assesses the stability of the current market structure.
- **TDFI (Time Decay Flow Imbalance):** Net flow into options highly sensitive to time decay.
- **TDPI (Time Decay Pressure Indicator):** Measures market impact of time decay, especially near expiration.
- **VFI (Volatility Flow Imbalance):** Net flow into options sensitive to volatility changes.

- **VRI (Volatility Risk Indicator):** Quantifies risks/opportunities from changes in implied/realized volatility.
- **Directional Signal:** Bullish/Bearish/Neutral signal based on MSPI and SAI.
- **SDAG Conviction Signal:** Bullish/Bearish signal based on agreement among multiple SDAG methodologies.
- **Volatility Expansion Signal:** Indicates conditions favoring an increase in market volatility.
- **Volatility Contraction Signal:** Indicates conditions favoring a decrease in market volatility.
- **Time Decay Pin Risk Signal:** High probability of price pinning to a strike due to time decay.
- **Time Decay Charm Cascade Signal:** Potential for accelerated moves due to charm-induced delta decay hedging.
- **Complex Structure Change Signal:** Warning that the current market structure may be shifting.
- **Complex Flow Divergence Signal:** Alert for divergences between price and cumulative order flow.

X. Appendix

(This section can be used for future additions, such as detailed mathematical formulas if desired by the user, or links to external resources.)